

ALEX SALAMATI

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EDUCATION

Loyola Marymount University, Los Angeles, CA

Expected May 2027

B.S. Computer Engineering, Minor: Computer Science

Relevant Coursework: Computer Architecture (SystemVerilog), Digital Systems, Microprocessors/Microcontrollers, Circuits I & II, Electronics, Data Structures & Algorithms

Fall 2026: Embedded Systems (Xilinx Zynq SoC, AXI, HLS), Graduate Machine Learning, Senior Capstone

TECHNICAL SKILLS

Hardware Description: Verilog, SystemVerilog

EDA & Tools: Xilinx Vivado (synthesis, simulation, timing analysis), Multisim, MPLAB X, Git/GitHub, Linux, Oscilloscopes, Logic Analyzers

Languages: C, C++, Python, MATLAB, Bash, MIPS Assembly

Concepts: RTL Design, FSM Design, Systolic Arrays, INT8/INT4 Quantization, Directed Testbench Verification, Static Timing Analysis, Setup/Hold Constraints, Logic Synthesis, CMOS Analog Design

SELECTED HARDWARE PROJECTS

Folded, Quantized Systolic MAC Array Accelerator | Senior Capstone | Verilog | OpenLane | sky130

Sep 2026 – Present

- Own RTL and architecture on a 3-person team building a small AI accelerator: a systolic multiply-accumulate (MAC) array for quantized matrix multiplication, the core operation of GPU tensor cores
- Designing the processing-element datapath (INT8 baseline, INT4 variant) and a folding-controller FSM that time-multiplexes a small physical PE array to emulate a larger logical array, trading latency for area
- Parameterizing fold factor and bit-width for a design-space sweep of area, power, timing slack, and accuracy; flow spans Python golden-model verification, Xilinx FPGA bring-up, and OpenLane RTL-to-GDSII on SkyWater sky130 with DRC/LVS signoff

MIPS Single-Cycle Processor RTL Design | SystemVerilog | FPGA | Vivado

Mar 2026 – May 2026

- Designed a complete 32-bit single-cycle MIPS processor including a custom ALU, control unit, register file, and full datapath, written from scratch in SystemVerilog
- Synthesized on a Xilinx FPGA in Vivado; reviewed logic resource utilization (LUTs, flip-flops) and analyzed critical path delays
- Built directed testbenches across R-type, I-type, and J-type instructions; ran RTL simulation and timing analysis, verifying setup/hold constraints across 15+ instructions

Custom Two-Stage CMOS Op-Amp | Analog IC Design | Multisim | Frequency Compensation

Feb 2026 – Mar 2026

- Designed a transistor-level two-stage CMOS operational amplifier achieving >60 dB open-loop gain with low input offset voltage
- Applied Miller compensation to secure >45° phase margin; validated stability via Bode plots and AC/DC sweep simulation

RESEARCH EXPERIENCE

Undergraduate Researcher, CubeSat Communications Subsystem | LMU | Advisor: Dr. Gustavo Vejarano

May 2026 – Jul 2026

- Developed satellite-side firmware (C, Salvo RTOS on dsPIC33) for an RF jamming-detection CubeSat; owned the hardware-software interface between the Raspberry Pi payload, onboard computer, and UHF transceiver
- Verified end-to-end AX.25 downlink at 435 MHz to a GNU Radio ground station on USRP SDR hardware, and validated the uplink command path that triggers an on-demand payload poll and downlink
- Designed binary-semaphore I2C bus arbitration that eliminated recurring system freezes; root cause was the semaphore being released between 16-byte chunks, letting concurrent RTOS tasks corrupt messages mid-transfer
- Ran a byte-survival study confirming a 97-byte register ceiling and 0x00 truncation through the C-string interface, establishing mandatory hex-encoding for image data; documented the stack in a 20-page handoff manual

ADDITIONAL PROJECTS

Ball Balancing Robot | PID Control | OpenCV | Arduino

Mar 2026 – Present

- Designing a closed-loop 3-DOF servo platform; OpenCV vision (30+ FPS) feeds a PID controller driving servo PWM, tuned with step-response analysis to minimize overshoot

SO-ARM101 Robot Arm | Servo Bus Control | LeRobot | 3D Printing

Apr 2026 – Present

- Assembled and calibrated a 6-DOF robot arm with serial servo bus control; achieved leader-follower teleoperation through LeRobot and am training an ACT imitation-learning policy for pick-and-place

EXPERIENCE

IT Support Technician | LMU ITS Classroom Support | Work Study

Jun 2024 – Present

- Debugged cross-platform hardware and OS issues across 20+ smart classrooms; documented recurring failure patterns and escalated firmware-level faults to vendors
- Diagnosed and resolved AV, OS, and hardware failures in minutes during live lectures under strict uptime requirements

LEADERSHIP & ACTIVITIES

Vice President, Additive Manufacturing Club | LMU

2024 – Present

- Led CAD review and 3D print process optimization; taught members design-for-manufacturing across FDM and resin systems

Member, Robotics Optimization & Building Organization (ROBO) and IEEE Student Chapter | LMU

2023 – Present

- Collaborated on robotics builds with sensors, motors, and embedded control; attended IEEE workshops on PCB design and DSP